

CyberSOC Platform

General Availability (GA)

Deployment Playbook

Enterprise Production Cutover Procedure
Djezzy Algeria - ANRT Compliant SOC Operations

Document Version: 2.0.0-GA

Classification: CONFIDENTIAL - Internal Use Only

Effective Date: 2026-08-26

Review Cycle: Before Each Production Deployment

Owner: CyberSOC DevOps Team

Approvals Required: CTO, Security Lead, Network Ops Manager

CRITICAL: This document contains sensitive operational procedures for the CyberSOC Platform. Unauthorized distribution or disclosure is strictly prohibited. All personnel involved in the GA deployment must read and acknowledge this playbook before proceeding with any production changes.

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1. Executive Summary

This Deployment Playbook provides the definitive procedure for transitioning the CyberSOC Platform from staging to General Availability (GA) production status at Djezzy Algeria. The platform represents a critical national security infrastructure asset, requiring meticulous planning, coordination, and execution to ensure zero-downtime cutover while maintaining strict ANRT compliance throughout the deployment process.

1.1 Scope and Objectives

The GA deployment encompasses the complete CyberSOC ecosystem including the core security operations center platform, SS7/telecom fraud detection systems, threat intelligence integration modules, compliance monitoring frameworks, and all supporting infrastructure components. The primary objectives of this deployment are to establish a fully operational, enterprise-grade SOC capability that meets or exceeds all regulatory requirements set forth by the Algerian National Regulatory Authority for Telecommunications (ANRT), while simultaneously delivering measurable improvements in threat detection capabilities, incident response times, and overall security posture visibility for Djezzy's network infrastructure and subscriber base.

1.2 Key Success Criteria

Criterion	Target Metric	Measurement Method
Zero Data Loss	100% data integrity	Pre/post migration checksum validation
Maximum Downtime	< 15 minutes total	Application uptime monitoring
Performance Baseline	< 5% degradation vs staging	Load testing comparison
Security Compliance	100% checklist completion	Security team sign-off
Rollback Readiness	< 30 minutes to revert	Rollback drill execution time

1.3 Risk Assessment Overview

The deployment carries inherent risks that have been identified, assessed, and mitigated through careful planning. The highest-risk activities include database schema migration where data transformation occurs, DNS cutover which affects external connectivity, and certificate rotation which impacts TLS termination. Each risk has been assigned an owner, mitigation strategy, and contingency plan that are documented in detail within this playbook. The overall risk posture for this deployment has been assessed as MODERATE with appropriate controls in place to reduce likelihood and impact of potential failure scenarios.

2. Pre-Deployment Checklist

Every item in this checklist MUST be completed and verified before proceeding with the GA deployment. No exceptions are permitted without explicit written approval from both the CTO and Security Lead. Each checklist item requires sign-off from the responsible party, and completed checklists must be archived for audit purposes as part of the organization's compliance documentation requirements.

2.1 Infrastructure Prerequisites

Item	Status	Verified By	Date/Time
Kubernetes cluster (9+ nodes) operational	☐		
PostgreSQL HA cluster (3 primary + 2 replicas)	☐		
PgBouncer connection pooling configured	☐		
Kafka cluster (9 brokers) running	☐		
Elasticsearch cluster (Hot/Warm/Cold tiers)	☐		
Redis Sentinel (HA cache) deployed	☐		
NVMe SSD storage provisioned	☐		
Network policies (Zero Trust) applied	☐		
RBAC permissions validated	☐		
Backup systems tested and verified	☐		
Disaster recovery site synchronized	☐		

2.2 Security Requirements

Security verification is non-negotiable for GA deployment. The following security-specific items must be validated by the Security Team lead before any deployment activities commence. Any security finding classified as HIGH or CRITICAL must be remediated and re-verified prior to proceeding.

Security Check	Requirement	Status	Evidence
TLS Certificates	Valid Let's Encrypt certs for *.djezzy.dz	☐	Certificate output
.env.production	All secrets generated (no defaults)	☐	Secrets audit log
Feature Flags	ALLOW_MFA_BYPASS=false confirmed	☐	DB query result
Rate Limiting	Global limits active (500/min)	☐	Config verification
CSP Headers	Strict policy applied	☐	Header scan output
Audit Logging	7-year retention enabled	☐	Log config dump
Database SSL	sslmode=require enforced	☐	Connection string

RBAC Roles	Least privilege verified	☐	Role matrix review
Network Policies	Zero Trust rules active	☐	kubect! output
Vulnerability Scan	No CRITICAL/HIGH findings	☐	Scan report ref

3. Deployment Phases

The GA deployment is executed in five distinct phases, each with specific timelines, responsibilities, and success criteria. Phases are designed to be sequential but allow for parallel execution of certain tasks where dependencies permit. Each phase includes defined checkpoints that must be passed before proceeding to the next phase. The entire deployment timeline spans approximately 30 hours from initiation to full production operation, with the majority of work completed during off-peak hours to minimize end-user impact.

3.1 Phase 1: Database Migration (T-24h to T-18h)

Database migration represents the most critical and time-sensitive phase of the entire GA deployment. During this phase, we migrate the complete CyberSOC data model from SQLite (development/staging) to PostgreSQL High Availability cluster (production). This migration includes schema transformation, data type conversion, index creation, partition setup, and comprehensive data integrity validation. The migration script has been extensively tested against staging data and achieves sub-second downtime through careful use of online schema migration techniques where supported by PostgreSQL.

3.1.1 Migration Commands

```
# Execute PostareSQL schema creation
psql -h aa-primarv.cybersoc.svc -U cybersoc app -d cybersoc production \
-f scripts/database/aa-migration-staging.sql

# Run data migration (SQLite → PostareSQL)
./scripts/database/execute-staging-migration.sh

# Validate migration integrity
python3 scripts/database/validate-aa-migration.py --verbose

# Verify row counts match source
psql -c "SELECT COUNT(*) FROM users;"
psql -c "SELECT COUNT(*) FROM incidents;"
psql -c "SELECT COUNT(*) FROM alerts;"
```

3.1.2 Migration Validation Checklist

Validation Step	Expected Result	Actual	Pass/Fail
Schema created successfully	All 25 tables present		
Indexes built	48 indexes created		
Enums registered	11 enum types present		
Users migrated	Count matches source DB		
Audit log preserved	Full history intact		
Foreign keys valid	No orphaned records		
Feature flags correct	Security flags = false		
Connection test	App can connect/query		

3.2 Phase 2: TLS Certificate Setup (T-12h to T-6h)

Transport Layer Security certificates are essential for protecting all platform communications in production. This phase involves requesting and validating certificates from Let's Encrypt via cert-manager, configuring internal service mesh mTLS certificates for inter-service communication, and setting up automatic renewal monitoring to prevent certificate expiration incidents. The certificate setup must be completed well in advance of the main cutover to allow for DNS propagation and ACME challenge completion, which can take variable amounts of time depending on network conditions and CA load.

3.2.1 Certificate Installation

```
# Apply cert-manager resources
kubectl apply -f k8s/cert-manager/certificates.yaml

# Verify certificate issuance (may take several minutes)
kubectl get certificates -n cybersoc -w

# Check certificate details once Ready
kubectl describe certificate soc-diezzv-dz-tls -n cybersoc

# Validate TLS secret exists
kubectl get secret soc-djezzy-dz-tls-prod -n cybersoc
```

3.3 Phase 3: Application Deployment (T-6h to T-2h)

Application deployment encompasses the complete containerized stack including the Next.js frontend API layer, backend microservices, background job processors, and all supporting components. This phase uses Helm charts for declarative deployment to Kubernetes, ensuring reproducible and auditable deployments. The deployment follows a canary release pattern initially routing 5% of traffic to validate the new version before progressively increasing traffic share based on error rate and latency metrics.

```
# Deploy using Helm with production values
helm upgrade --install soc-platform ./k8s/helm/soc-platform \
-n cybersoc \
-f k8s/helm/soc-platform/values-production.yaml \
--wait --timeout 600s

# Verify pod status
kubectl get pods -n cybersoc -l app.kubernetes.io/name=soc-platform

# Check Horizontal Pod Autoscaler status
kubectl get hpa -n cybersoc

# Verify Ingress configuration
kubectl get ingress -n cybersoc
```

3.4 Phase 4: Monitoring & Validation (T-2h to T-0h)

The monitoring and validation phase establishes comprehensive observability for the production environment and executes final pre-go-live checks. This phase deploys Grafana dashboards customized for the Djezzy SOC operations context, configures Prometheus alerting rules aligned with SLA targets, sets up PagerDuty and Slack notification integrations for incident response, and conducts thorough end-to-end testing of all critical user journeys and system integrations.

3.4.1 Dashboard Import

```
# Import Grafana dashboards
./scripts/import-grafana-dashboards.sh \
--url https://grafana.soc.diezzv.dz \
--api-key ${GRAFANA_API_KEY}

# Verify Prometheus datasource
curl -s ${GRAFANA_URL}/api/datasources \
-H "Authorization: Bearer ${GRAFANA_API_KEY}" | jq '.[]'

# Test alerting rules
promtool test rules monitoring/prometheus/soc_alerts.yml
```

3.5 Phase 5: Cutover Execution (Go-Live)

CRITICAL PHASE: The cutover execution represents the point of no return for the GA deployment. Once initiated, rollback becomes increasingly complex as real user data begins flowing through the system. This phase requires all hands on deck with clear communication channels established between all teams. The Go/No-Go decision must be made unanimously by the deployment committee based on Phase 1-4 results.

Time	Action	Owner	Status
T-0:00	Final Go/No-Go decision meeting	Deployment Lead	[]
T+0:05	Announce maintenance window start	Communications	[]
T+0:10	DNS cutover to production ingress	Network Ops	[]
T+0:15	Verify HTTPS connectivity	Security Team	[]
T+0:20	Enable user authentication	Platform Team	[]
T+0:25	Activate real-time alerting	SOC Analyst	[]
T+0:30	Begin canary traffic (5%)	DevOps	[]
T+0:45	Check error rates & latency	SRE Team	[]
T+1:00	Increase to 25% traffic	DevOps	[]
T+1:15	Validate business transactions	QA Team	[]
T+1:30	Increase to 50% traffic	DevOps	[]
T+1:45	Monitor system under load	SRE Team	[]
T+2:00	Full traffic cutover (100%)	Deployment Lead	[]

T+2:15	End maintenance mode announcement	Communications	[]
T+2:30	Post-cutover validation complete	All Teams	[]

4. Rollback Procedures

Despite extensive preparation and testing, production deployments carry inherent risk. This section defines the rollback procedures for each deployment phase, enabling rapid restoration of the previous stable state if critical issues are detected during or after the cutover. Rollback decisions should be made quickly based on predefined criteria rather than attempting to diagnose issues in production. The guiding principle is: when in doubt, roll back first and investigate later in a controlled staging environment.

4.1 Immediate Rollback Triggers

The following conditions require IMMEDIATE rollback without waiting for further investigation or management approval. These triggers represent situations where continued operation poses unacceptable risk to data integrity, system security, or user experience. Any team member observing these conditions is authorized and expected to initiate rollback procedures immediately.

Trigger Condition	Severity	Response Time	Auto-Rollback?
Data corruption detected	CRITICAL	< 1 minute	Yes
Authentication bypass possible	CRITICAL	< 1 minute	Yes
Error rate > 10%	HIGH	< 5 minutes	Yes
P95 latency > 5 seconds	HIGH	< 5 minutes	Manual
Database connection failures	HIGH	< 2 minutes	Yes
Certificate expiration imminent	MEDIUM	< 1 hour	Manual
SS7 integration failures	HIGH	< 5 minutes	Manual
Compliance violation detected	CRITICAL	< 1 minute	Yes

4.2 Rollback Commands

```
# DATABASE ROLLBACK (if migration failed)
# Point application back to old SQLite/PostgreSQL
export DATABASE_URL=
# Restart application pods
kubectl rollout restart deployment/soc-platform-api -n cybersoc

# APPLICATION ROLLBACK (Helm)
# Revert to previous release version
helm rollback soc-platform -n cybersoc

# DNS ROLLBACK
# Update DNS to point back to staging
# Via cloud provider console or API

# EMERGENCY FULL ROLLBACK
# Scale down production, scale up staging
```

```
kubectl scale deployment soc-platform -n cybersoc --replicas=0
```

```
# Verify staging is handling traffic
```

```
# VERIFY ROLLBACK SUCCESS
```

```
kubectl get pods -n cybersoc
```

```
kubectl rollout status deployment/soc-platform -n cybersoc
```

5. Post-Deployment Verification

Following successful cutover, a systematic verification process confirms all systems are operating correctly under production load. This verification goes beyond simple health checks to include functional testing of business-critical workflows, performance benchmarking against established baselines, security validation of production configurations, and integration testing with all dependent services. The verification process typically requires 2-4 hours depending on the complexity of issues discovered and resolved.

5.1 Functional Verification Matrix

Function Area	Test Case	Priority	Result	Notes
Authentication	User login (LDAP/SAML)	P0	[PASS/FAIL]	
Authentication	MFA enforcement	P0	[PASS/FAIL]	
Incident Mgmt	Create incident	P0	[PASS/FAIL]	
Incident Mgmt	Assign & escalate	P0	[PASS/FAIL]	
Alerting	Alert ingestion	P0	[PASS/FAIL]	
Alerting	Correlation rules	P1	[PASS/FAIL]	
Threat Intel	IoC lookup	P1	[PASS/FAIL]	
Threat Intel	MISP sync	P1	[PASS/FAIL]	
SS7 Monitoring	Message capture	P0	[PASS/FAIL]	
SS7 Monitoring	Fraud detection	P0	[PASS/FAIL]	
Reporting	Dashboard load	P1	[PASS/FAIL]	
Reporting	Export functionality	P2	[PASS/FAIL]	
Compliance	Audit log writing	P0	[PASS/FAIL]	
Compliance	ANRT gateway submit	P0	[PASS/FAIL]	

5.2 Performance Benchmarks

Performance measurements must be compared against baseline values captured during staging environment testing. Any deviation greater than 10% from baseline requires investigation and explanation. Performance degradation beyond 25% from baseline may warrant rollback pending optimization. Key metrics to monitor include API response latency at various percentiles (P50, P95, P99), database query execution times, message queue processing lag, and browser-side page load times for the dashboard interface.

Metric	Staging Baseline	Production Target	Measured	Status
API P50 latency	< 100ms	< 120ms		
API P95 latency	< 500ms	< 750ms		
API P99 latency	< 2000ms	< 3000ms		

DB query avg	< 50ms	< 75ms		
Page load (dashboard)	< 3s	< 4s		
SS7 msg processing	< 100msg/s	< 150msg/s		
Concurrent users	100 sustained	500 sustained		
Error rate	< 0.1%	< 0.5%		

6. Hypercare Period Guidelines

The hypercare period extends for 72 hours following the GA cutover, during which enhanced monitoring, accelerated incident response, and continuous presence of key personnel ensures rapid resolution of any issues that may surface under real production workload. This period represents the highest state of operational readiness, with all hands available for immediate response to any anomaly detected across the platform. The hypercare period concludes only after 72 hours of stable operation without critical incidents and with all key performance indicators remaining within acceptable thresholds.

6.1 Hypercare Schedule

Period	Coverage	On-Call Rotation	Response SLA
Hours 0-24	24/7 coverage	All teams on standby	5 min P1, 15 min P2
Hours 24-48	Extended hours	Core team + on-call	10 min P1, 30 min P2
Hours 48-72	Business hours+	On-call rotation	15 min P1, 1 hour P2
Post-72h	Standard ops	Normal rotation	Per SLA

6.2 Hypercare Checkpoints

Four formal checkpoint meetings occur during the hypercare period to assess system health, address accumulated issues, and determine readiness to transition to standard operations. Each checkpoint produces a go/no-go decision for proceeding to the next phase, with specific criteria that must be met. Checkpoint attendees include representatives from Development, Operations, Security, and Business stakeholder groups to ensure cross-functional awareness and alignment on system status.

Checkpoint	Timing	Key Decision Criteria	Decision Maker
CP1	+6 hours post-cutover	No P1 incidents, error rate < 1%	Deployment Lead
CP2	+24 hours post-cutover	All P0 tests passing, latency OK	Engineering Manager
CP3	+48 hours post-cutover	User feedback positive, stability good	Product Owner
CP4	+72 hours post-cutover	Ready for standard ops handoff	CTO

7. Emergency Contacts & Escalation

Clear escalation paths are essential for effective incident response during the deployment and hypercare periods. The contact information below should be verified current before each deployment cycle and updated as personnel changes occur. All team members should have these contacts saved in personal devices and accessible even if corporate systems are unavailable. Consider printing physical copies for the war room during deployment.

7.1 Primary Escalation Contacts

Role	Name	Phone	Email	Availability
Deployment Lead	[Name]	+213 XX XXX XXXX	@djezzy.dz	24/7 during deploy
CTO	[Name]	+213 XX XXX XXXX	@djezzy.dz	Emergency only
Security Lead	[Name]	+213 XX XXX XXXX	@djezzy.dz	24/7 during deploy
Network Ops Mgr	[Name]	+213 XX XXX XXXX	@djezzy.dz	Business hours + on-call
DBA On-Call	[Name]	+213 XX XXX XXXX	@djezzy.dz	24/7 rotation
SRE On-Call	[Name]	+213 XX XXX XXXX	@djezzy.dz	24/7 rotation
ANRT Liaison	[Name]	+213 XX XXX XXXX	@anrt.dz	Business hours

7.2 External Vendor Contacts

Service	Vendor	Support Contact	Contract #	SLA
Cloud Infrastructure	[Provider]	[Phone/Email]	#CONTRACT-001	99.9% uptime
CDN/DDoS Protection	Cloudflare	[Phone/Portal]	#CONTRACT-002	< 5min response
Certificate Authority	Let's Encrypt	Community Support	N/A	Best effort
Monitoring	Grafana Labs	[Support Portal]	#CONTRACT-003	P1: 15min
Identity Provider	[IdP Vendor]	[Support Portal]	#CONTRACT-004	P1: 30min